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Molecular approaches for detecting encapsulated bacteria

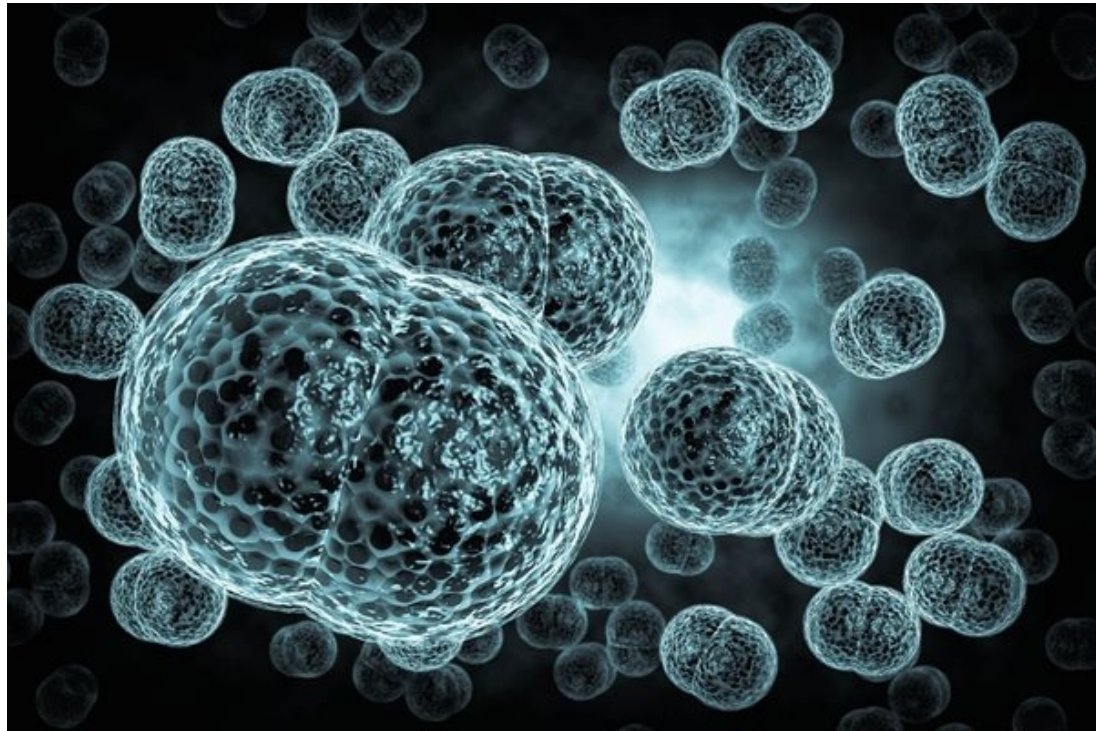
Dam Khan

Tiemele Laurent Simon Amoikon

Noutin Fernand Michodigni

Brenda Kwambana Adams

14/09/2025



Meningitis burden

- Meningitis is an infection of the protective membranes that surround the brain and spinal cord (meninges).
- Every year, there are an estimated 2.3 million cases of meningitis around the world (<https://tracker.meningitis.org/cases/>)
- Up to 83% of meningitis cases worldwide occur in low or lower-middle income countries.
- According to WHO, the four main causes of acute bacterial meningitis are *Neisseria meningitidis* (meningococcus), *Streptococcus pneumoniae* (pneumococcus), *Haemophilus influenzae*, and *Streptococcus agalactiae* (group B streptococcus).



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Cerebrospinal fluid (CSF) as the diagnostic sample

- The collection of CSF is an invasive procedure and is performed by experienced personnel under aseptic conditions.
- Collected CSF samples are transported to microbiology laboratory and plated within 1 hour for culture results
- TI medium can be used to keep organisms viable during transport if culture within 1 hour is not possible.
- CSF specimens can also be stored at -70C for PCR

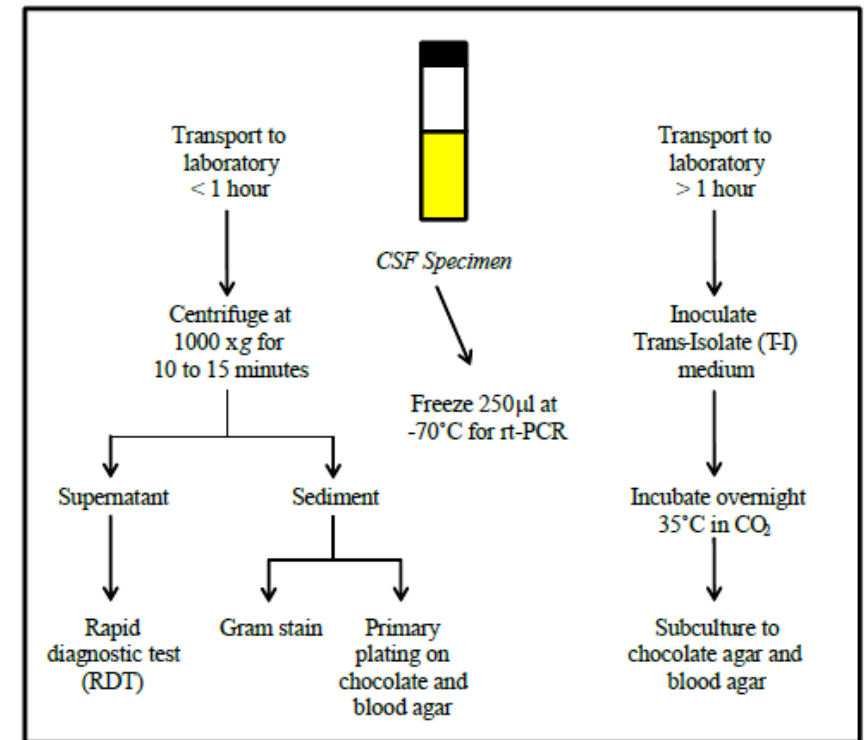


Fig: Processing of CSF samples.
source: WHO manual for diagnosis of meningitis pathogens

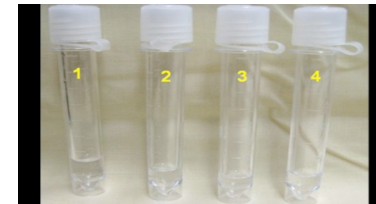
Presumptive identification of meningitis pathogens

- Macroscopic examination of CSF

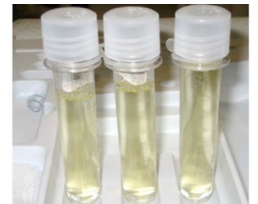
Turbidity of CSF is associated with bacterial meningitis

- Cytological examination of CSF

- WBC counts >10 cells/mm³
- decrease glucose concentration (<45 mg/dl)
- and increased protein concentration (>45 mg/dl) are associated with bacterial meningitis



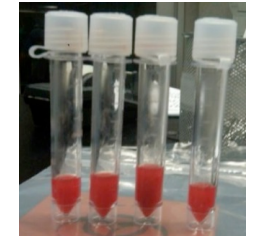
Clear



Xanthochromic



Turbid



Bloody

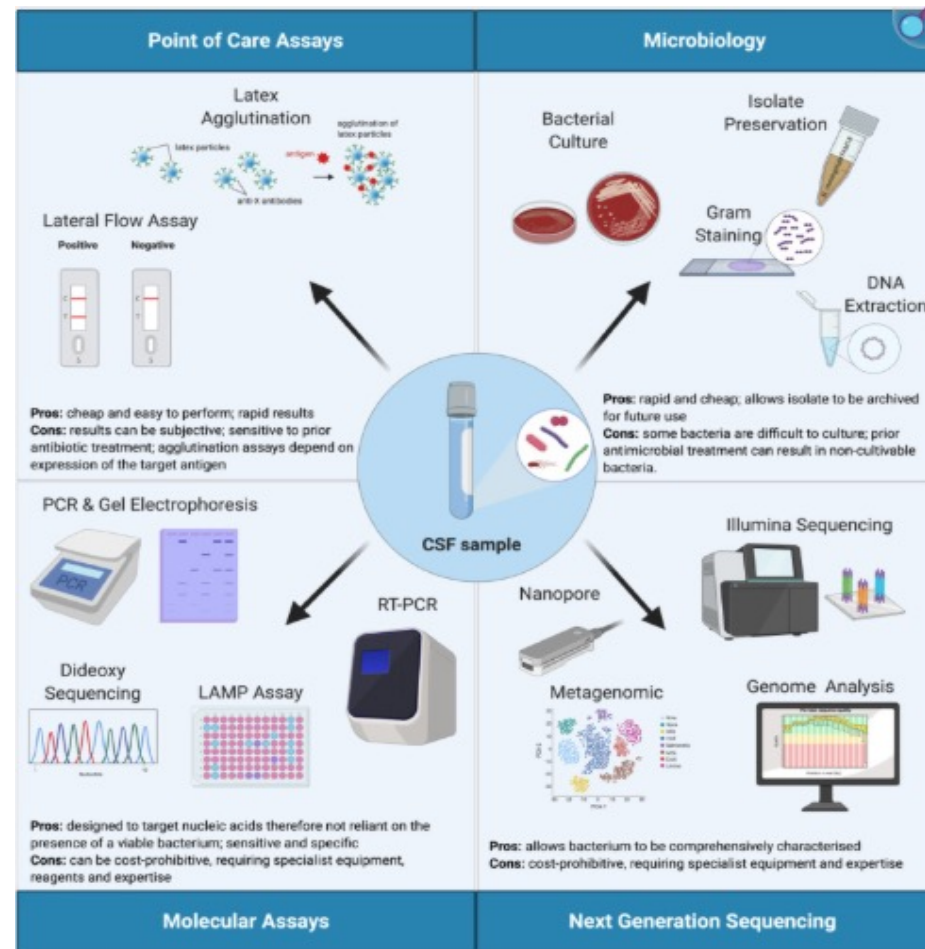


Fig: Approaches used in the diagnostics of meningitis pathogens. Source (Kanny Diallo et al. 2021)

Rapid diagnostic tests

Any test which can be used in healthcare settings with little infrastructure or trained personnel, preferably without electricity.

❖ Latex agglutination

- Pastorex Meningitis Kit/(Wellcogen)
- BD Directigen Combo Kit

❖ Immunochromatography

- BinaxNOW *S. pneumoniae* Kit (Alere)
- MeningoSpeed™ (BSD_0200)



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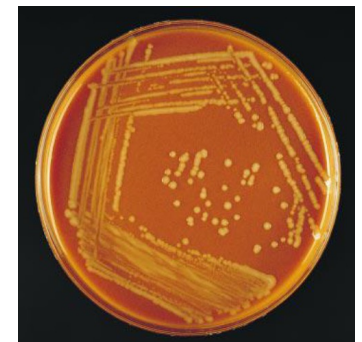


Microbiology culture

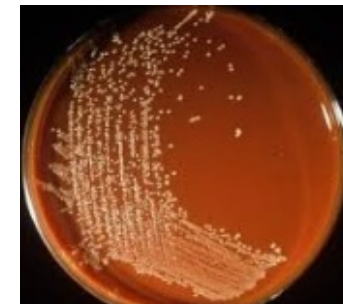
- CSF transported within an hour from the time of collection or stored in TI media can be inoculated to obtain a primary culture
- Positive rate by culture is relatively low due to suboptimal storage and transportation conditions, culture practice, and/or antibiotic treatment administered before the specimen is collected
- Presumptive identification of *S. pneumoniae*, *N. meningitidis* and *H. influenzae* can be obtained based on growth in culture plate and gram stain results.
- Biochemical tests can be used for the definitive identification of the three bacteria



S. pneumoniae



H. influenzae



N. meningitidis

Polymerase chain reaction (PCR)

- ❖ Polymerase Chain reaction (PCR): A technique used in molecular biology to amplify a single copy or a few copies of a segment of DNA across several orders of magnitude, generating millions of copies of a particular DNA sequence
- ❖ Arguably the most powerful laboratory technique ever invented
 - Easy to perform
 - Low cost
 - Combination of specificity and sensitivity
 - Flexible



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Short history of PCR

- ❖ 1983: Kary Mulis developed PCR
- ❖ 1985: First publication of PCR by Cetus corporation appears in Science
- ❖ 1986: Purified Taq polymerase is first used in PCR
- ❖ 1988: PerKinElmer introduces the automated thermal cycler
- ❖ 1989: Science declares Taq polymerase “molecule of the year”
- ❖ 1993: Kary Mulis shared Nobel prize in Chemistry for conceiving PCR technology

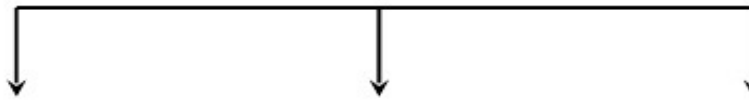


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Applications of PCR



Molecular Identification

- Molecular Archaeology
- Molecular Epidemiology
- Molecular Ecology
- DNA fingerprinting
- Classification of organisms
- Genotyping
- Pre-natal diagnosis
- Mutation screening
- Drug discovery
- Genetic matching
- Detection of pathogens

Sequencing

- Bioinformatics
- Genomic cloning
- Human Genome Project

Genetic Engineering

- Site-directed mutagenesis
- Gene expression studies



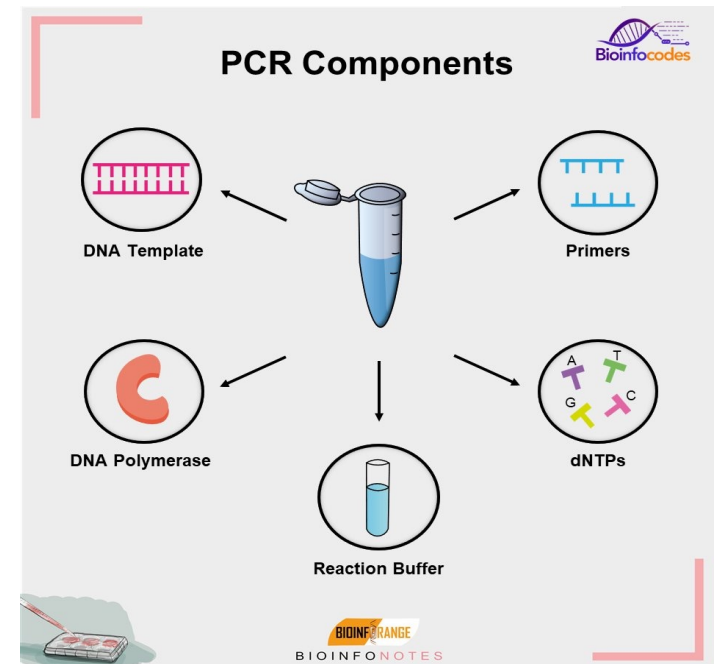
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Components of a PCR

- Target DNA
- deoxynucleotide triphosphate (dNTPs)
- $MgCl_2$
- Forward Primer
- Reverse Primer
- Polymerase
- Water
- Thermal cycler



Steps of setting up a PCR

Prepare and clean workspace and equipment

Prepare mastermix and add template

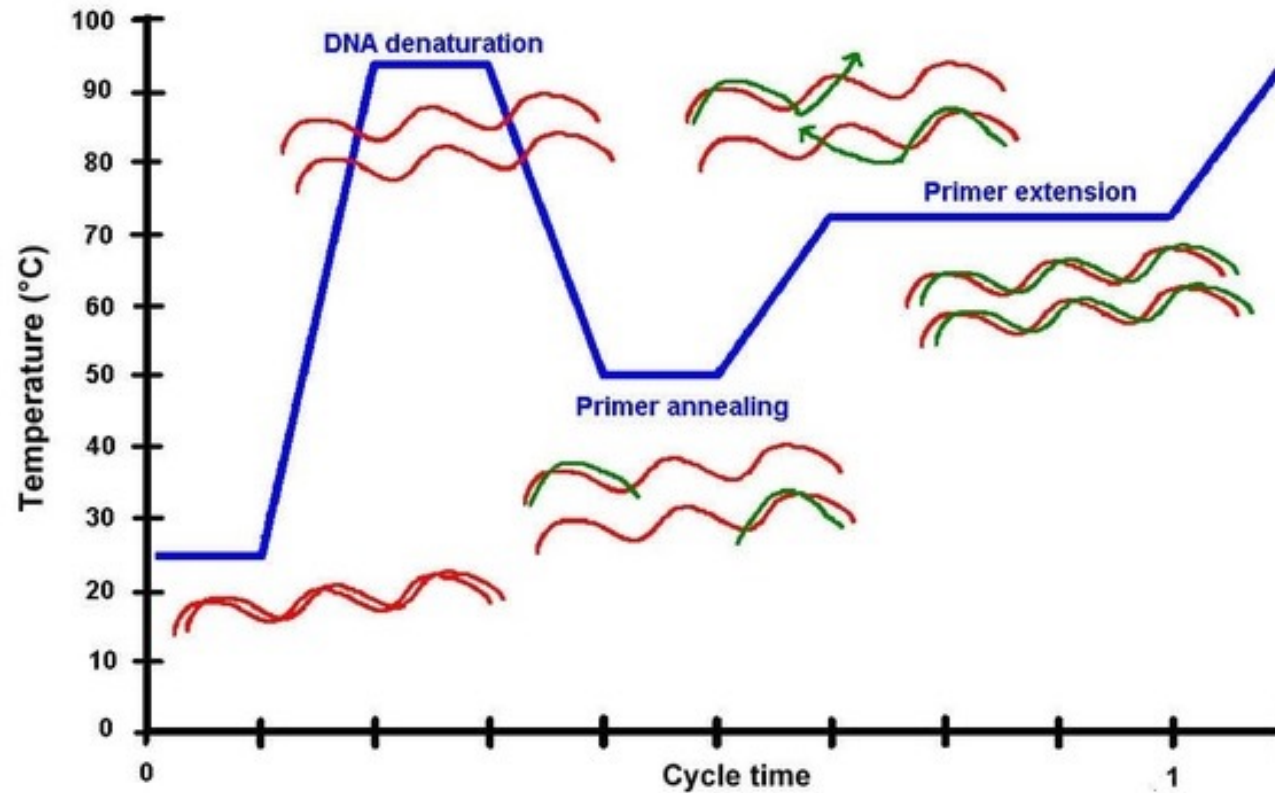
Amplify DNA in a thermocycler

Check amplification products

Clean and maintain workspace



Typical Profile of a PCR Reaction



Conventional PCRs

- ❖ Detects products at the end point of DNA amplification by visualizing amplicons using gel electrophoresis
- ❖ Gel-based detection requires that tubes containing PCR amplicons be opened and manipulated
 - Greatly increases the risk of contamination of laboratory space, equipment, and reagents with amplified materials
 - Very time consuming, less sensitive than real-time PCR
 - Primarily used for typing assays employing purified cultures or clinical specimens that contain the organism in high density



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Viewing the PCR products

❖ Agarose gel electrophoresis

Steps:

1. Prepare the Agarose Gel
2. Cast the Gel
3. Load the DNA Samples
4. Run the Gel
5. Separation of DNA Fragments
6. Visualization
7. Interpreting the Results

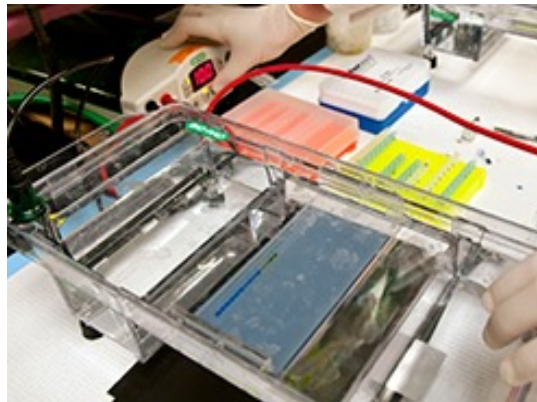
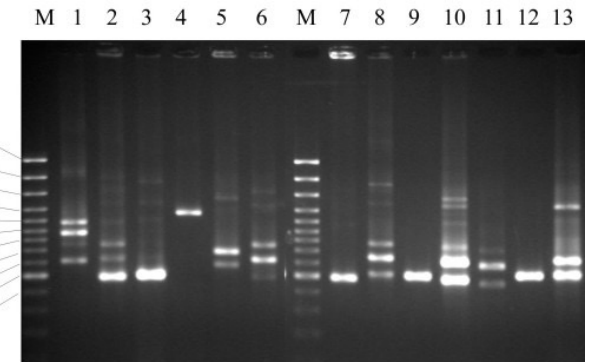


Figure 2

A

3000 bp
2000 bp
1500 bp
1200 bp
1031 bp
900 bp
800 bp
700 bp
600 bp
500 bp
400 bp



❖ Automated electrophoresis QIAXCEL



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Real time PCR

- ❖ Real time PCR allows you to monitor the progress of the PCR reaction in real-time, providing quantitative data about the initial amount of the target nucleic acid in the sample.
- ❖ Specific detection systems rely on fluorescent resonance energy transfer (FRET) probes
 - Specifically recognize target sequences
 - Systems of choice for the molecular detection assays
 - More expensive than non-specific detection systems
- ❖ Dual- labelled hydrolysis probe - An oligonucleotide (~17-35 bp long) labeled with a reporter fluorophore at the 5' end and a quencher fluorophore at the 3' end

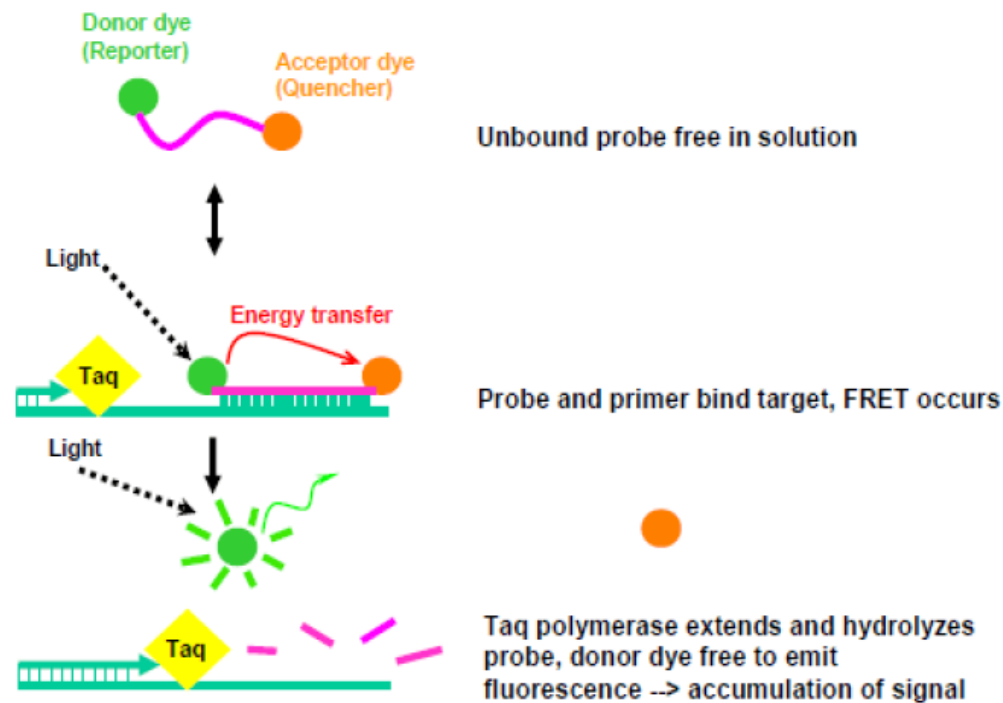


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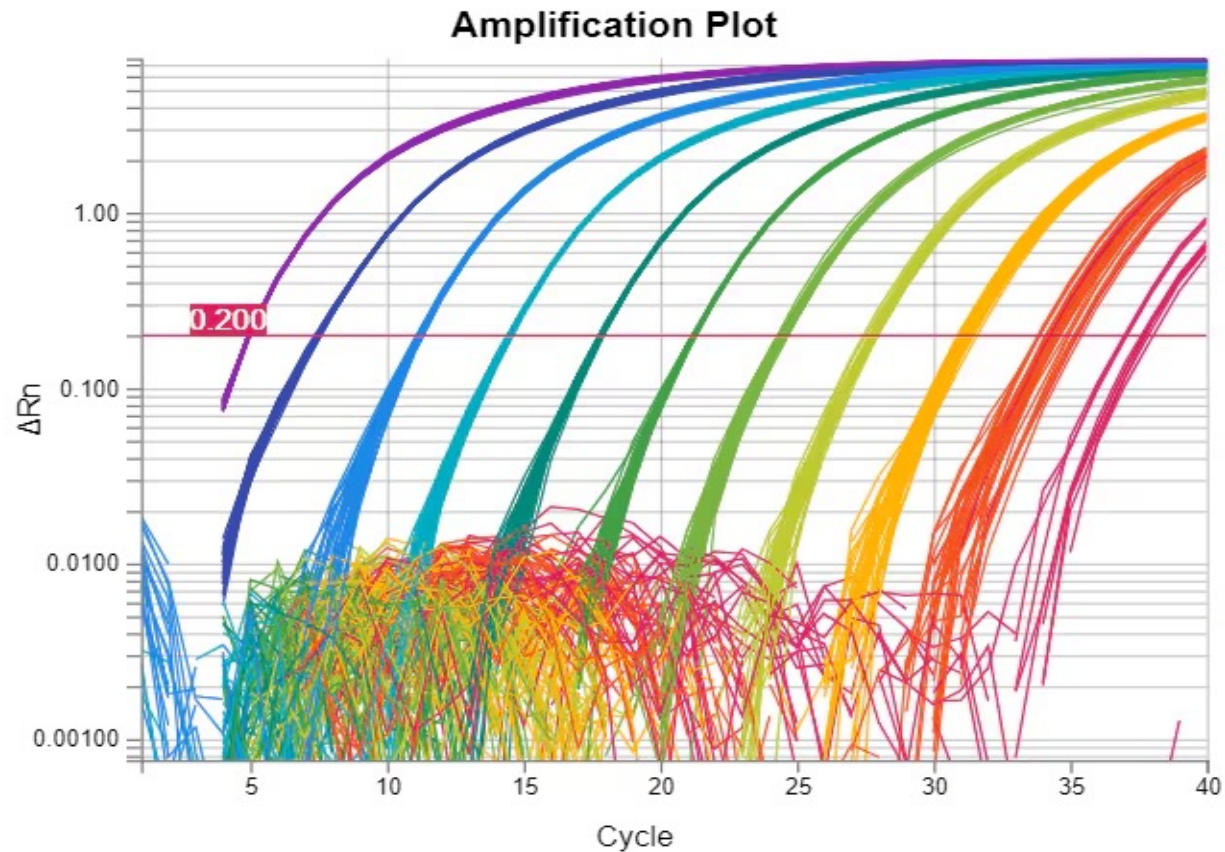


Chemistry of a dual-labeled hydrolysis probes



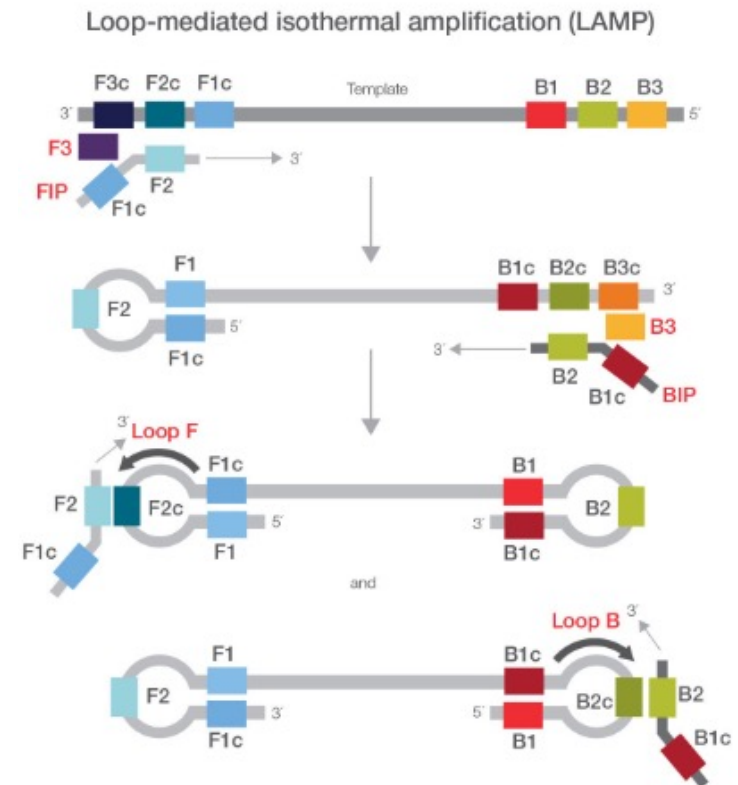
A fluorescent signal is only generated if the probe interacts with its specific target and is subsequently hydrolyzed during amplification. The resulting increase in fluorescence is proportional to the amount of amplified PCR product in the reaction

Amplification plots generated by a real-time PCR assay



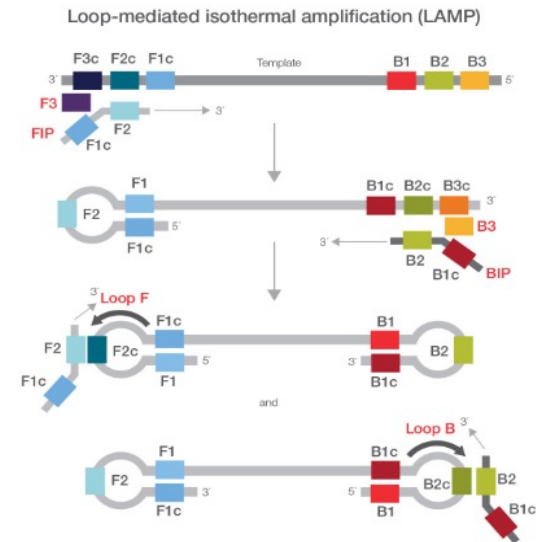
loop-mediated isothermal amplification (LAMP)

- A LAMP assay is a rapid nucleic acid amplification method used for detecting specific DNA or RNA targets
- LAMP assays for the detection of *S. pneumoniae*, *N. meningitidis* and *H. influenzae* type b have been developed (Kanny Diallo et al. 2021)



Chemistry of LAMP

- Primers: use of 5-6 primers of which 2 inner primer (FIP and BIP)
- Enzyme:
 - ✓ DNA polymerase with strong strand displacement activity
 - ✓ No thermal cycling needed, constant incubation (60-65°C)
- Amplification
 - ✓ DNA synthesis with strand displacement
 - ✓ Formation of stern-loop DNA structures
 - ✓ Cyclic amplification generates cauliflower-like DNA with multiple loops



Comparaison LAMP vs PCR

Feature	PCR	LAMP
Equipment	Expensive thermocycler	Simple heater or portable device
Time to result	~2 h	~1 h 20 min
Cost per test	~20-25 USD	~10-15 USD
Sensitivity to inhibitors	High	Low
Specificity	Good (2-3 primers)	Very high (5-6 primers)
Field use	Limited	Suitable (portable, minimal training)

Source : Tumino et al., 2020



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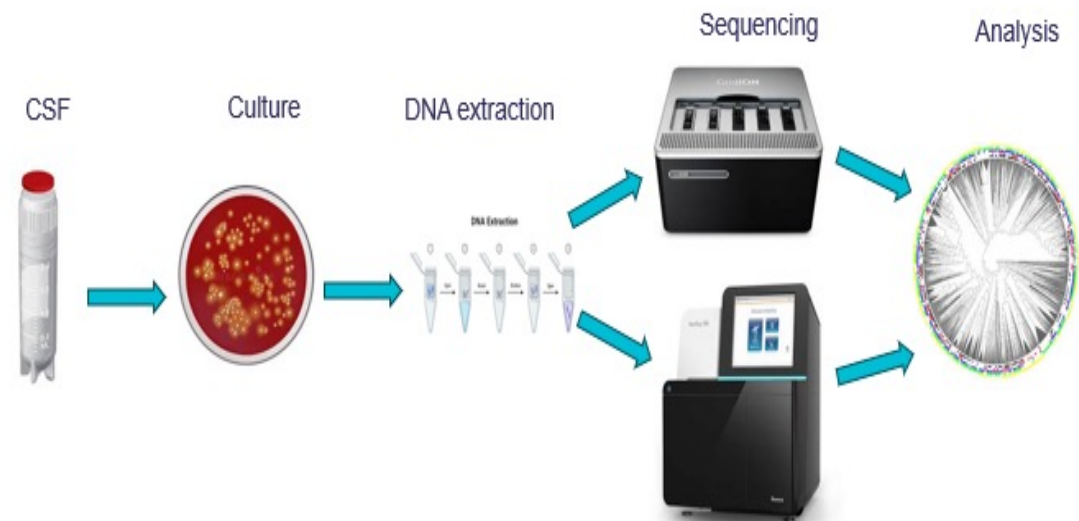
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Whole-Genome Sequencing (WGS) of isolates

WGS of isolates provides genetic information which can be used for:

- Accurate species identification
- Serotyping / strain typing
- Detection of antimicrobial resistance genes
- Virulence factor profiling
- Outbreak investigation & global surveillance



Metagenomics sequencing

- Metagenomics is the sequencing of microbial DNA directly from clinical or environmental samples.
- Detects organisms present without prior culture.
- Two main approaches: 16S rRNA gene sequencing and Shotgun metagenomics.



Bioinformatics tools for characterisation of encapsulated bacteria post sequencing

- Species identification: Kraken2
- Serotyping: *S. pneumoniae* : SeroBA
N. meningitidis: meningotype
H. influenzae: hicap
- MLST: PubMLST
- Antimicrobial Resistance Prediction: ResFinder, CARD
- Virulence Factor detection: VFDB
- Phylogenetics & Outbreak Tracking: Snippy, Gubbins



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Application of PCR in the detection and characterization of bacterial meningitis pathogens at the WHO RRL in The Gambia



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IBD surveillance in West and Central Africa

- WHO RRL supports 17 sentinel sites in 12 West and Central African countries
- All the countries have introduced PCV and Hib
- WHO RRL provides technical support during outbreaks in the West Africa

Country	Year of Hib Introduction	Year of PCV Introduction
Benin	2005	2011
Burkina Faso	2006	2013
Cameroon	2009	2011
Cote d'Ivoire	2009	2014
The Gambia	1997	2009
Ghana	2002	2012
Mali	2007	2011
Niger	2008	2014
Nigeria	2012	2014
Senegal	2005	2013
Sierra Leone	2007	2011
Togo	2008	2014

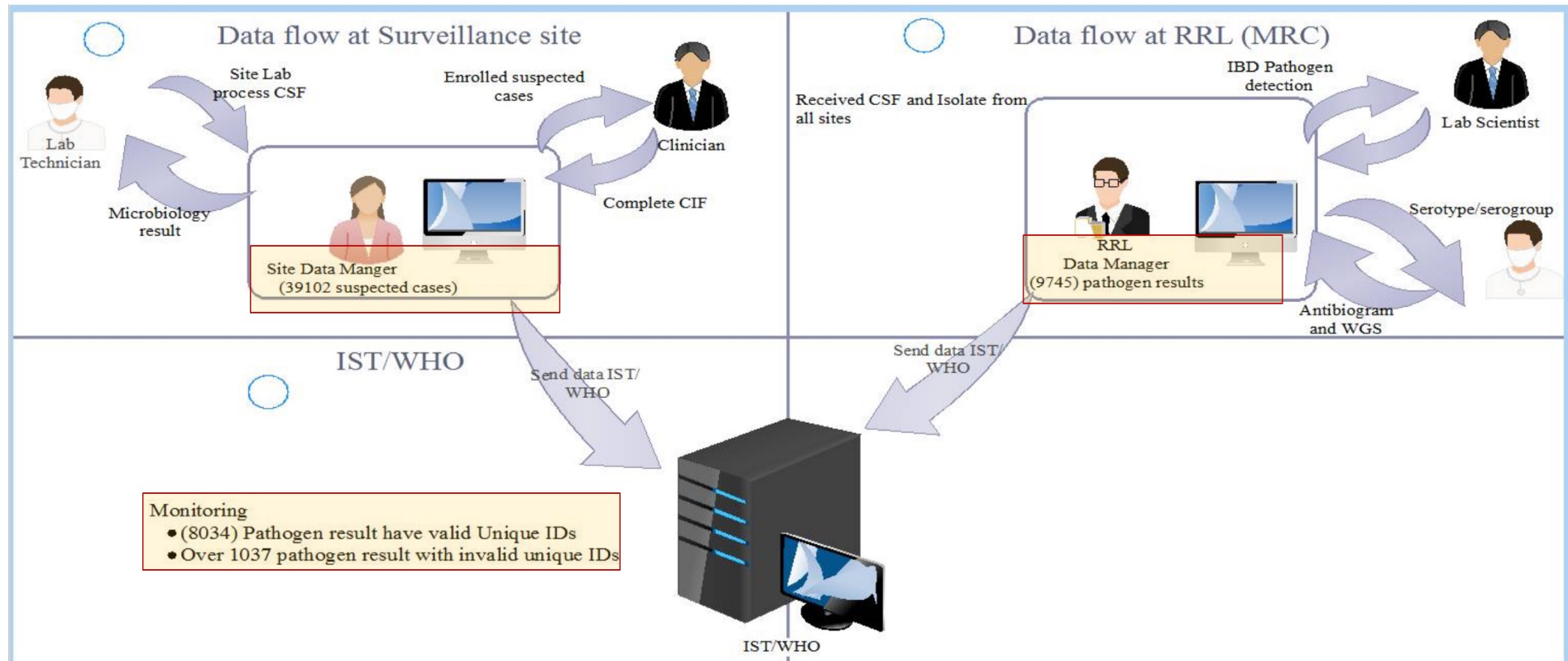


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Workflow at the sentinel sites and reference lab



Laboratory tests performed at the WHO Reference lab in The Gambia

- ❖ Isolates from the PBM sentinel sites are sent to the WHO RRL for further characterization:
 - Confirmation of isolate identification
 - Serotyping/serogrouping (latex, block PCR/qPCR)
 - Antimicrobial susceptibility testing
 - Whole genome sequencing (MLST, resistome,...)

- ❖ Aliquots of CSF from suspected cases including the culture negative ones are also sent to the WHO RRL in the Gambia where molecular methods are used to detect and further characterize potential bacterial meningitis pathogens that could have been missed out during culture.

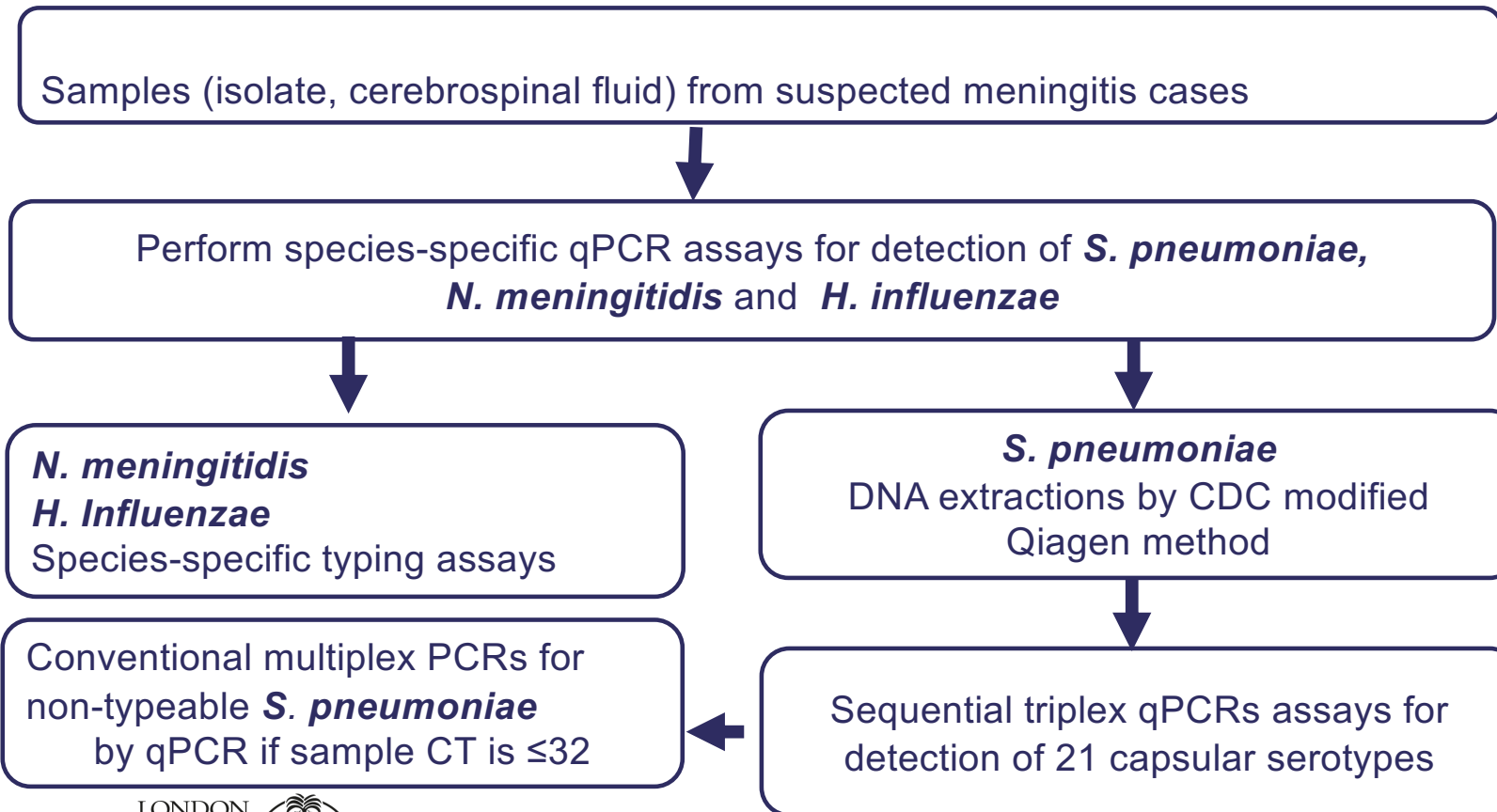


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Workflow for detection and serotyping of bacterial meningitis pathogens by real time PCR



Pathogen detection

real-time PCR

Organism	Gene target (WHO recommend targets)	Gene targets (in press, Simon, Kanny, Keith, James, Odile, Martin, Frontiers in Microbiology)
<i>Streptococcus pneumoniae</i>	<i>lytA</i>	<i>SP2020</i>
<i>Neisseria meningitidis</i>	<i>SodC</i>	<i>SodC</i>
<i>Haemophilus influenzae</i>	<i>hpd</i>	<i>dmsA</i>
<i>Streptococcus agalactiae</i> (not systematically tested)	-	<i>cfb</i>



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Pathogen detection

PCR equipment



Reagents and Chemicals

- PerfeCTa® qPCR ToughMix®
- Primers and probes
- Nuclease free water



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Serotyping/Serogrouping of *Haemophilus influenzae* and *Neisseria meningitidis*

Organism	Target gene
NmA	<i>SacB</i>
NmB	<i>SynD</i>
NmC	<i>SynE</i>
NmW	<i>SynG</i>
NmX	<i>xcbB</i>
NmY	<i>synF</i>

Organism	Target gene
Hia	<i>acsB</i>
Hib	<i>bcsB</i>
Hic	<i>ccsD</i>
Hid	<i>dcsE</i>
Hie	<i>ecsH</i>
Hif	<i>bexD</i>

Similar PCR conditions to the detection assay are used for the Serotyping/grouping of *N. meningitidis* and *H. influenzae*



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Serotypes targeted for the *S. pneumoniae* serotyping

21 targets

qPCR
1
5
23F
4
6A/6B/6C/6D
9V/9A
14
18C
19F
3
7F/7A
19A
6C/6D
12F/12A/12B/44/46
22F/22A
15A/15F
23A
33F/33A/37
2
11A/11D/11F
16F

Conventional PCR Serotypes	
14	2
1	11A/11D
5	19A
4	16F
18A/18B/18C/18F	21
6A/6B/6C	33F/33A/37
19F	15A/15F
23F	35F/47F
25F/38	13
9V/9A	39
7C/7B/40	23B
3	35A/35C/42
15B/15C	20
7F/7A	35B
17F	10F
8	34
12F/12A/44/46	10A
9N/9L	31
22F/22A	6C
23A	6A/6B/6C

40 targets



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Pediatric Bacterial Meningitis Surveillance

September 15 2019
Volume 69
Supplement 2



Clinical Infectious Diseases

Early Impact of Pneumococcal Conjugate Vaccines in African Countries

OXFORD
UNIVERSITY PRESS
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A Supplement to *Clinical Infectious Diseases*



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SUPPLEMENT

Volume 69, Issue Supplement_2, 15 September 2019

Early Impact of Pneumococcal Conjugate Vaccines in African Countries

SUPPLEMENT ARTICLES

Pediatric Bacterial Meningitis Surveillance in the World Health Organization African Region Using the Invasive Bacterial Vaccine-Preventable Disease Surveillance Network, 2011–2016

Jason M Mwenda and others

Clinical Infectious Diseases, Volume 69, Issue Supplement_2, 15 September 2019, Pages S49–S57,
<https://doi.org/10.1093/cid/ciz472>

We describe the epidemiology of meningitis cases due to *Streptococcus pneumoniae*, *Neisseria meningitidis*, and *Haemophilus influenzae* across 26 African countries participating in the World Health Organization's Invasive Bacterial Vaccine-preventable Diseases Network, including trends in vaccine-type pneumococcal meningitis post-pneumococcal conjugate vaccine introduction.

Abstract ▼ [View article](#) [Supplementary data](#)

Declines in Pneumonia and Meningitis Hospitalizations in Children Under 5 Years of Age After Introduction of 10-Valent Pneumococcal Conjugate Vaccine in Zambia, 2010–2016

Evans M Mpabalwani and others

Clinical Infectious Diseases, Volume 69, Issue Supplement_2, 15 September 2019, Pages S58–S65,
<https://doi.org/10.1093/cid/ciz456>

We evaluated the impact of the 2013 introduction of a 10-valent pneumococcal conjugate vaccine (PCV10), using hospital administrative data in Zambia. Substantial declines in pneumonia and meningitis hospitalizations among children aged <5 years were observed post-PCV10.

Abstract ▼ [View article](#)

Confirmed Pediatric Bacterial Meningitis

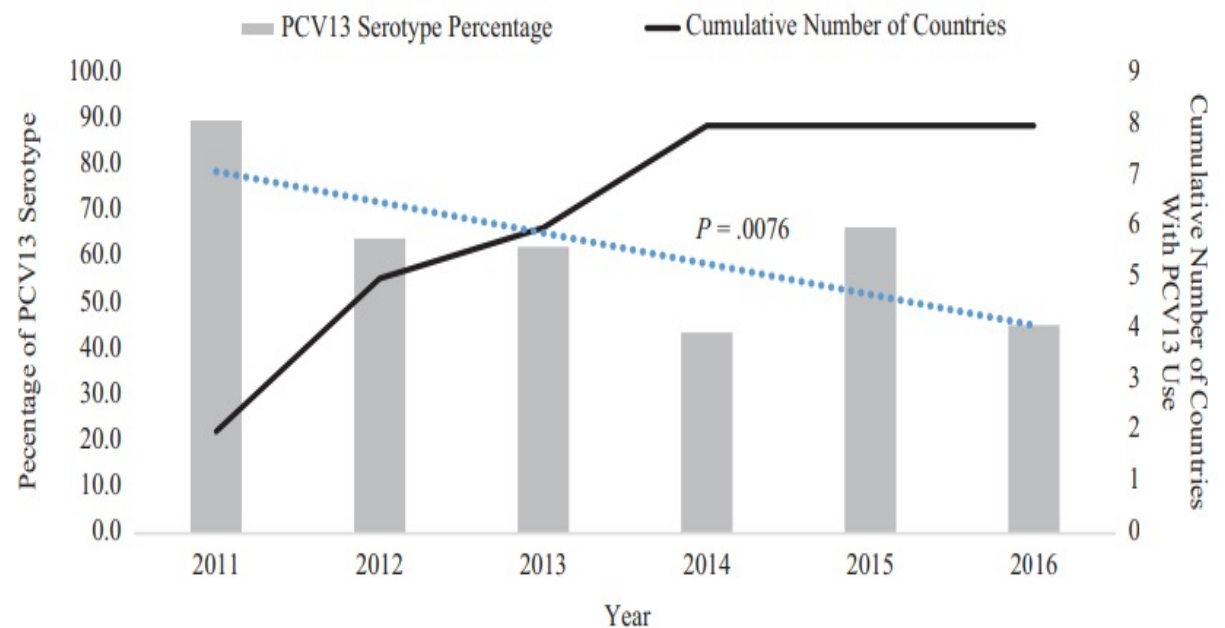
No/ of CSF specimens tested	No. of Confirmed cases	Confirmed Hi Cases (% of Confirmed Cases)	Confirmed Nm Cases (% of Confirmed Cases)	Confirmed Spn Cases (% of Confirmed Cases)
20 297	954	125 (13.1)	321 (33.6)	508 (53.2)

Findings

- Serogroup W was the most common *N. meningitidis* serogroup, 54.8% [34/62] in the West-Central Africa region
- Hib remained the most common serotype; 44.1% (19/43) among *H. influenzae* meningitis cases
- serotype 14 (n = 10 [10.8%]) was the most common causative agent *Streptococcus pneumoniae* meningitis

PCV introduction and annual proportion of confirmed *Streptococcus pneumoniae* meningitis cases due to vaccine serotypes

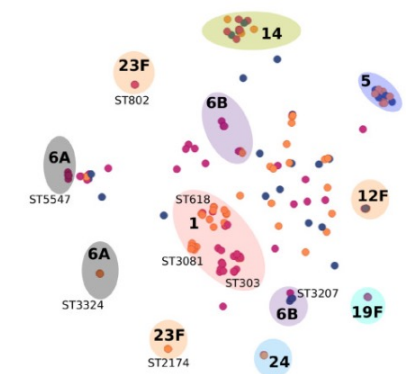
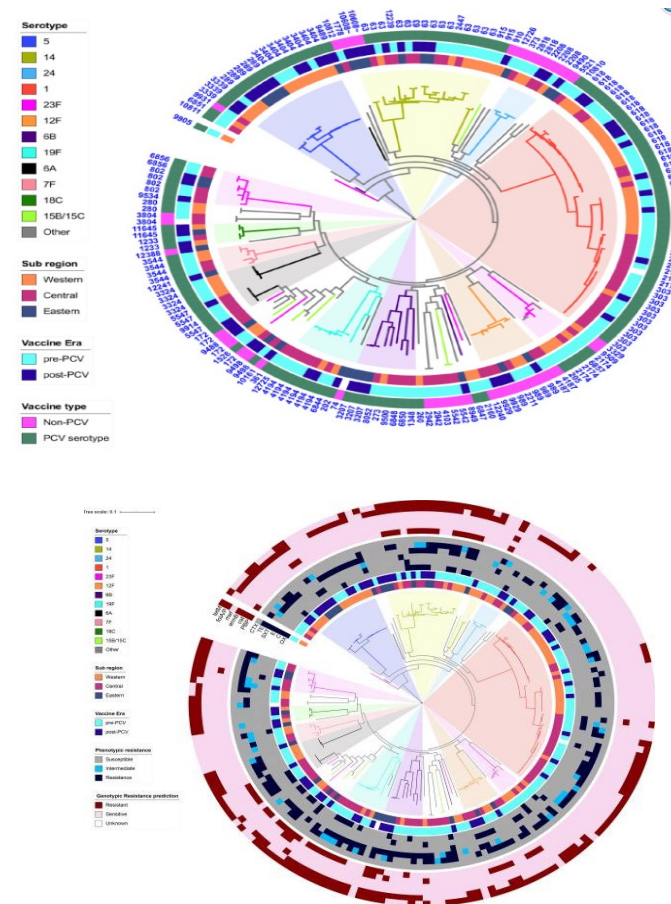
Country	Year of PCV introduction
Benin	2011
Cameroon	2011
Cote d'Ivoire	2014
Gambia	2011
Ghana	2012
Niger	2014
Nigeria	2014
Senegal	2013
Sierra Leone	2011
Togo	2014



Genomic characteristics of *S. pneumoniae* isolates collected during the surveillance

Findings

- Clustering and similarity of accessory genome content of Isolates from the same sub-region
- Antibiotic-resistance genotypes that confer resistance to penicillin, chloramphenicol, co-trimoxazole and tetracycline were detected across all sub-regions.



Madikay Senghore et al. 2021

Pneumococcal Outbreaks in West Africa

2016



2016

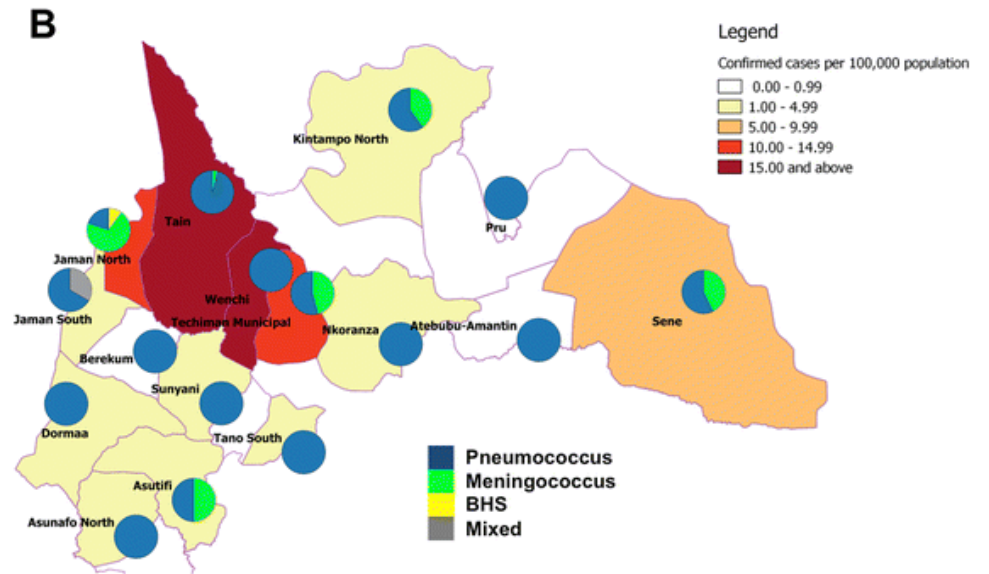
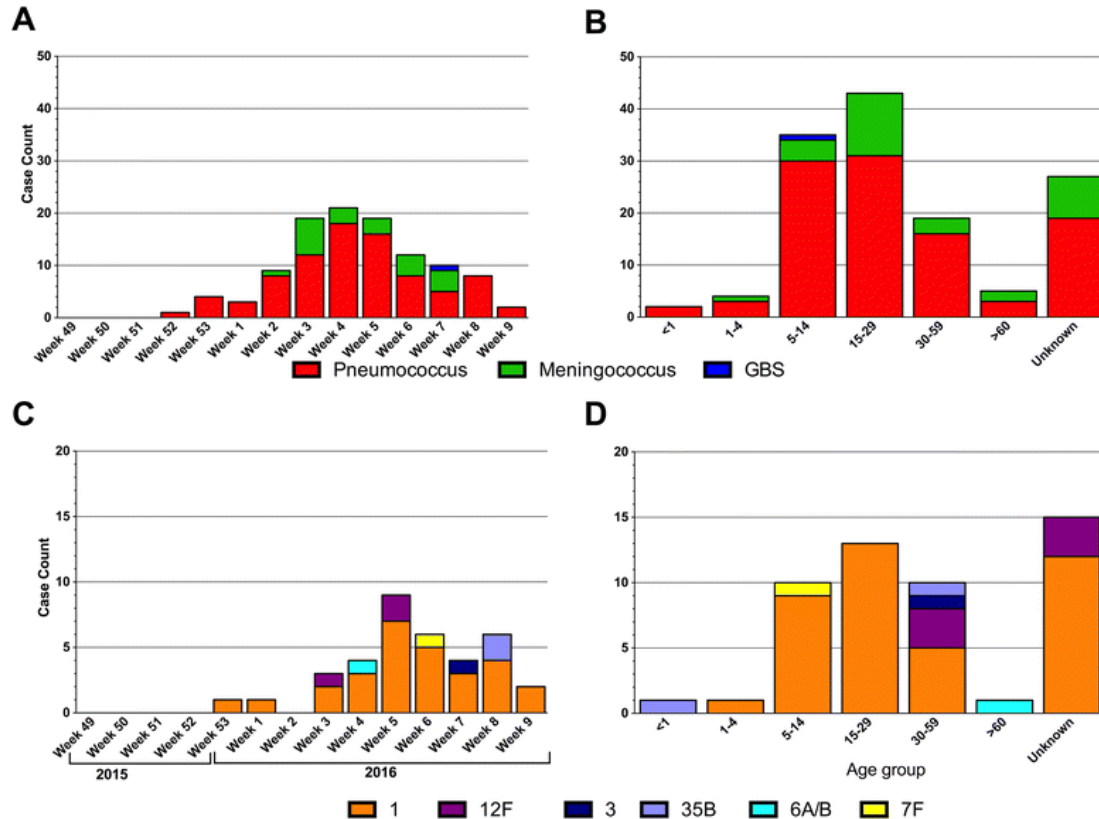


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Ghana 2016 pneumococcal meningitis outbreak



- Pneumococcal serotype 1 outbreak
- Most cases among older children and young adults
- Why didn't herd immunity protect unvaccinated older children and young adults?

Kwambana-Adams, BMC ID, 2016

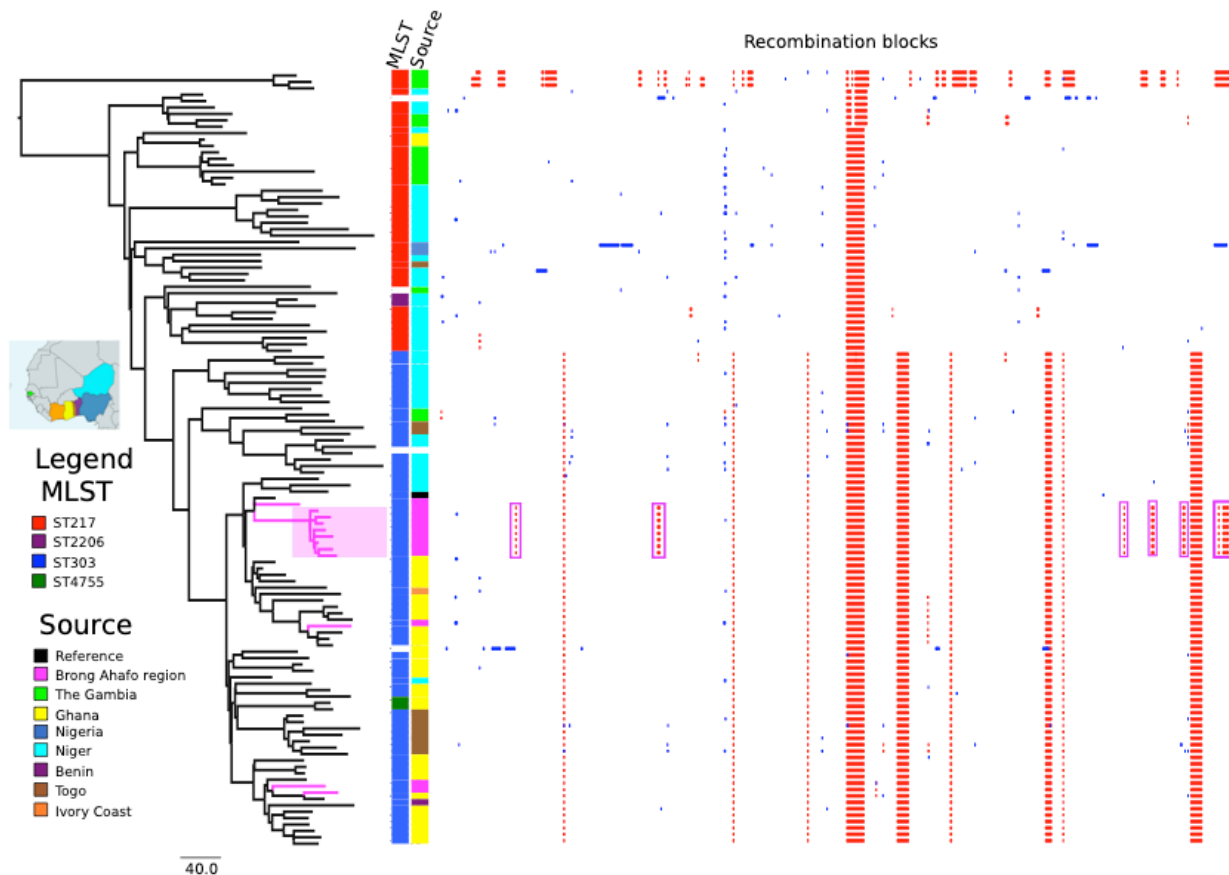


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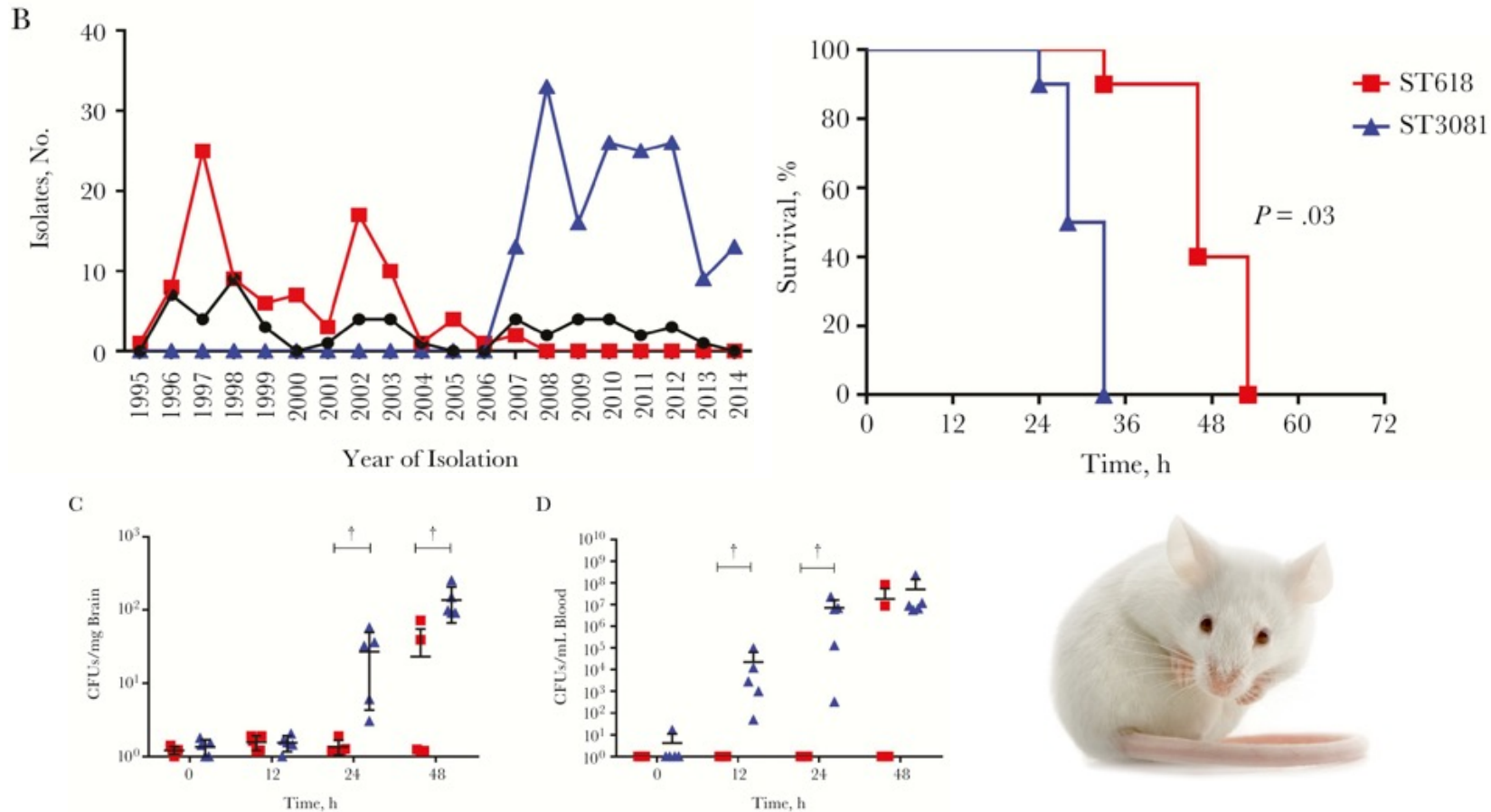
Genomic analysis of outbreak strain



In the novel clade, recombination affected regulatory genes

- *dltABCD* operon
- zinc uptake regulatory protein *acdA*
- MarR-family regulatory protein

Emergence of hypervirulent strain in The Gambia



Bricio-Moreno and Ebruke *et. al.*, Journal of Infectious Diseases, 2017



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Roll out of the NmA Vaccine

- Developed especially for Africa by Serum Institute India through the Meningitis Vaccine Project, coordinated by WHO and PATH
- Low-cost NmA conjugate vaccine (<USD1.00)
- 24/26 ABM countries conducted mass vaccination campaigns
- >340 million 1-29 year-olds vaccinated (2010-2020)
- Disappearance of NmA outbreaks

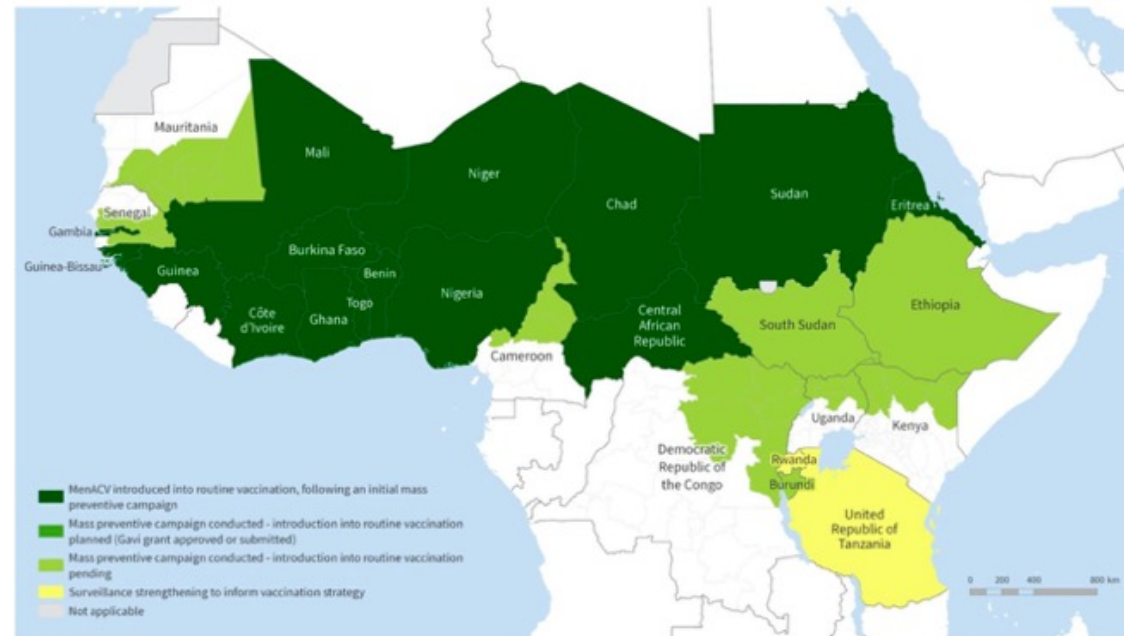


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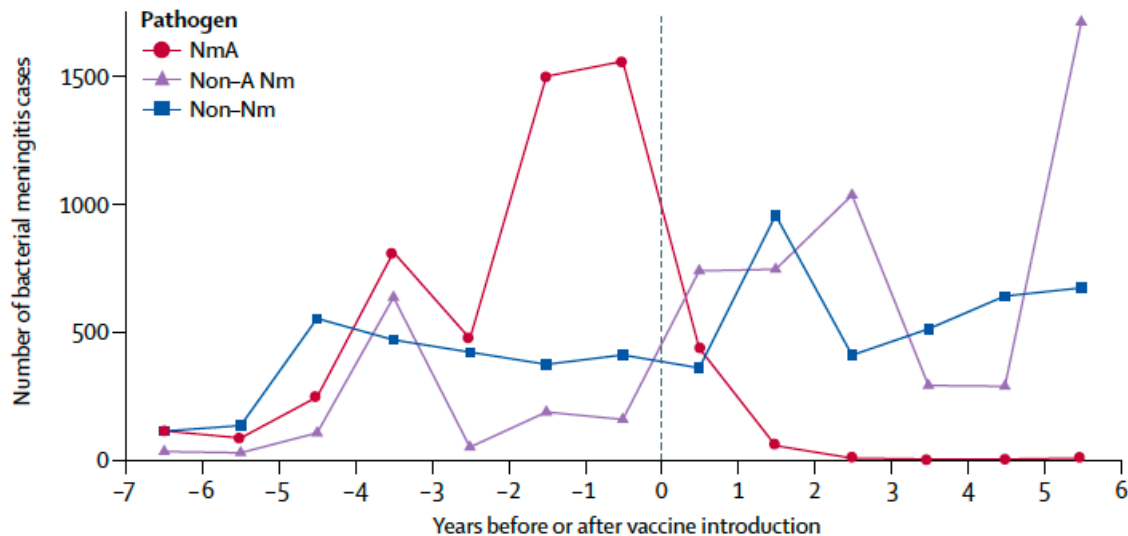


Status of meningococcal A conjugate vaccine (MenACV) roll out in the meningitis belt



WHO AFRO, Meningitis Bulletin, September 2023

Nm outbreaks before and after Nma vaccine



Trotter et al., Lancet Infect Dis, 2017

- Hypervirulent W, C and X strains with epidemic potential
- Atypical disease presentation

Non-A Nm Outbreaks (2011 – 2017)

Year	Serogroup	Sequence type(s)	Countries
2011	NmW	ST-11	Niger
2012	NmW	ST-11	Burkina Faso, The Gambia
2013	NmC	ST-10217	Nigeria
2014	NmC	ST-10217	Nigeria
2015	NmC	ST-103, ST-10217	Ethiopia, Niger, Nigeria
2016	NmW		Ghana
2017	NmC NmW	ST-10217, ST175 ST-11	Niger, Nigeria, Benin, Cameroun, Nigeria Togo

Adapted from Fernandez et al., JID, 2019



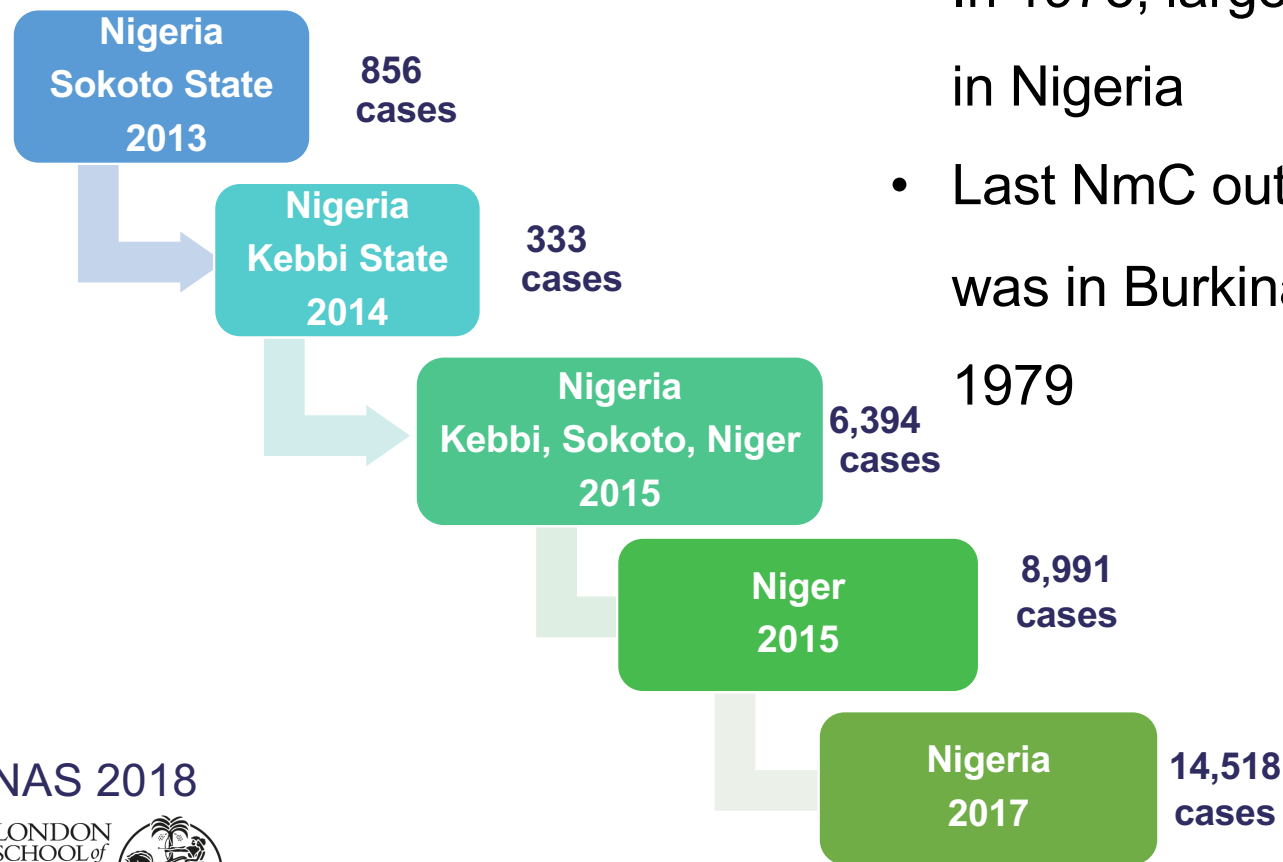
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NmC ST10217 in West Africa

- In 1975, large NmC epidemic in Nigeria
- Last NmC outbreak in Africa was in Burkina in Faso in 1979



Brynildsrud et al., PNAS 2018

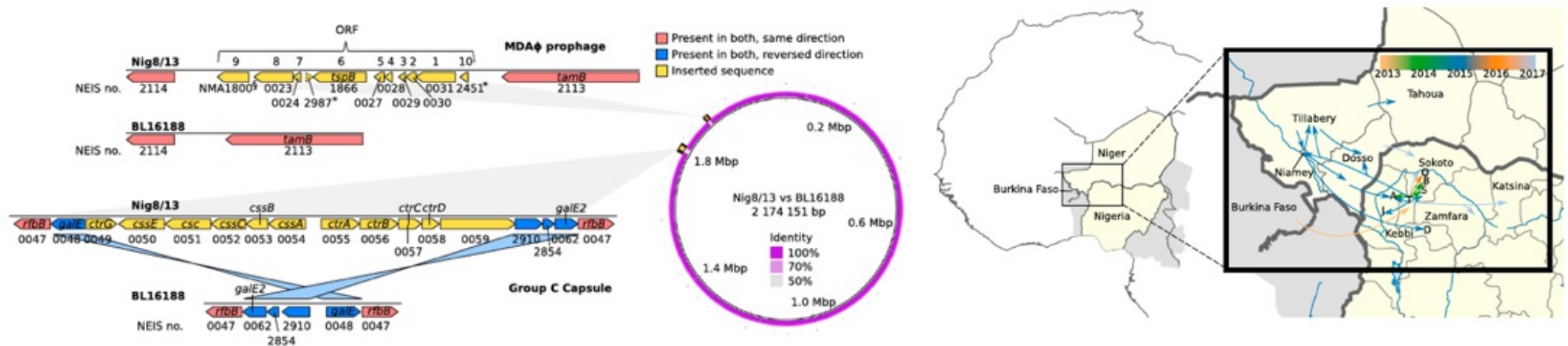


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Emergence of the hypervirulent NmC strain



- ❑ NMc ST-10217 Strain acquired serogroup C capsule genes
- ❑ MDA ϕ phage associated with increased virulence

Further reading

- 1) Molecular diagnostic assays for the detection of common bacterial meningitis pathogens: A narrative review. Kanny Diallo et al. 2021:
<https://pmc.ncbi.nlm.nih.gov/articles/PMC7957090/>
- 1) The paediatric meningitis routine surveillance:
https://academic.oup.com/cid/issue/69/Supplement_2
- 3) Nigeria *N. meningitidis* meningitis outbreak:
<https://www.nature.com/articles/s41598-018-32475-2>
- 4) Ghana *S. pneumoniae* meningitis outbreak:
<https://pubmed.ncbi.nlm.nih.gov/27756235/>



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Acknowledgements

Professor Martin Antonio

Dr Brenda Kwambana-Adams

Archibald Worwui

Catherine Okoi

Peter Sylvanus Ndow

Peggy-Estelle Tientcheu

Shola Able Thomas

Aissatou Seck

N'da Kouame Nazaire KOUADIO

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